

EEG Workshop - Hands-On Reference

BioRTC 2026 · Yobe State University EEG_01_Preprocessing.ipynb

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How to Use These Slides

These slides are **workshop reference material** - not a lecture.

- Each frame is labelled with its notebook section (Sec X) in the top-right corner
- The **orange accent** marks the lecture slides; **blue** marks these workshop slides
- Use these slides for quick reference and figures

Recording used throughout: aaaaakbz_s001_t000.edf (TUAR, 01_tcp_ar)

Two seizures: 439 s–505 s and 940 s–1058 s (cpsz, channel T3)

Sub-corpus	Label	What it contains	Used for
TUAR v3.0.1	artifact labels	eyem, musc, elec per channel	Notebook 1
TUSZ v2.0.6	seizure labels	onset/offset, 25k sessions	Notebook 2

Folder structure - use 01_tcp_ar throughout (average reference, >90% of recordings):

```
tuar/v3.0.1/edf/  
  01_tcp_ar/ ← use this  
  02_tcp_le/  
  03_tcp_ar_a/
```

File naming: aaaaakbz_s001_t000

aaaaakbz	patient ID (8 random letters)
s001	session 1 (1st hospital visit)
t000	token 0 (first segment)

TUAR artifact CSV (bname.csv)

Label	Meaning
eyem	eye movement / blink
musc	muscle (EMG) artifact
elec	electrode artifact
eyem_musc	eye + muscle overlap
musc_elec	muscle + electrode overlap
elpp	electrode pop (transient)

Format: no header row.

Columns: channel, start, stop, label, confidence

Read with: `pd.read_csv(path, comment='#', names=[...])`

TUSZ seizure CSV (bname.csv)

Label	Meaning
bckg	background (no seizure)
fnsz	focal non-specific seizure
cpsz	focal, impaired awareness
gnsz	generalized non-specific seizure
absz	absence (generalized, non-motor)
tcsz	tonic-clonic (generalized, motor)
tnsz	tonic (generalized, motor)

Format: has header row.

Read with: `pd.read_csv(path, comment='#')`

Binary task: any non-bckg label → seizure (1);
bckg → background (0).

Patient	Type(s)	Montage	Role
aaaaasgd	fnsz	01_tcp_ar	Main demo (EEG_02 / EEG_03)
aaaaampk	fnsz, gnsz	01_tcp_ar	
aaaaamnd	gnsz	01_tcp_ar	
aaaaaoys	gnsz	01_tcp_ar	
aaaaaict	cpsz	01_tcp_ar	added for type diversity
aaaaaoek	tcsz, tnsz	01_tcp_ar	added for type diversity
aaaaaazz	absz	02_tcp_le [†]	added for type diversity

All from TUSZ v2.0.6, dev split. [†] the only absz example available locally uses linear-ear montage, not 01_tcp_ar - adjust the loader path/channel suffix (-LE not -REF) accordingly.

EDF file (referential)

Channels stored as **electrode vs. reference**:

EEG FP1-REF

EEG F7-REF

EEG T3-REF

⋮

Annotation CSV (bipolar)

Artifacts annotated on **adjacent electrode pairs**:

FP1-F7

F7-T7

T7-P7

⋮

One channel per electrode. Ref: same for all.

Each label = difference between two ref channels.

You cannot directly match a CSV channel name to an EDF channel name

To map FP1-F7 $\rightarrow$ EDF: you need (EEG FP1-REF) - (EEG F7-REF) (construct the difference yourself).

Bandpass: 0.5 - 70 Hz

- `l_freq = 0.5`: removes DC drift & electrode movement
- `h_freq = 70`: removes high-frequency muscle noise

Notch

TUAR (US): `notch_filter(60)`

Nigeria: `notch_filter(50)`

Applying the wrong frequency leaves a residual spike that broadband classifiers will learn as a spurious feature.

Causal vs. zero-phase

MNE `filter()` default: **zero-phase**
(non-causal FIR)

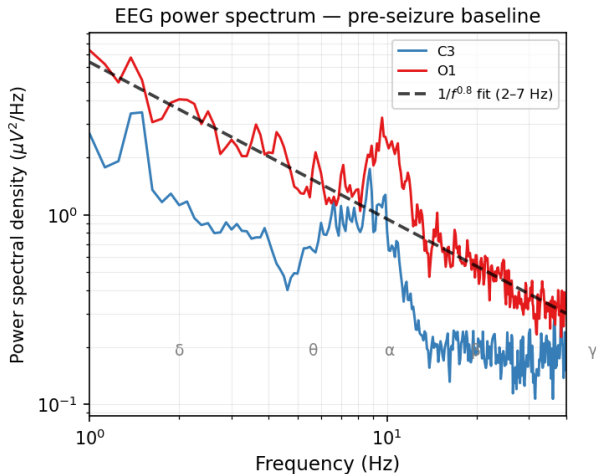
<i>Zero-phase</i>	<i>Causal (one-pass)</i>
No group delay	Phase delay = half filter length
Cannot be used in real-time	Required for on-line/BCI use
Standard for re-search	Standard for streaming

Important: a non-causal filter “knows the future” - never use it to evaluate online detection latency.

$$P(f) \propto \frac{1}{f^\alpha}, \quad \alpha \approx 1-2$$

- Universal in EEG - not pathological
- δ always dominates γ in absolute power
- Cross-band and cross-subject comparisons require normalization
- α rises during sleep ($\alpha \approx 2$) - confound for classifiers

Fix: baseline normalization (Section 6)



Channels C3 and O1, pre-seizure baseline.

Parameters in this notebook

nperseg = 8 s window length
noverlap = 7 s 1 s hop
window = 'hann' smooth edges

Frequency resolution

$$\Delta f = 1/\text{nperseg}$$

$$8 \text{ s window} \Rightarrow \Delta f = \mathbf{0.125 \text{ Hz}}$$

$$2 \text{ s window} \Rightarrow \Delta f = 0.5 \text{ Hz}$$

IEDs (~100-200 ms) are averaged out

An 8 s window spans the IED 40-80 $\times$ - the brief burst is diluted.

To resolve IEDs: shorten nperseg to ~ 0.5 s (sacrifice freq. resolution).

Heisenberg uncertainty

$$\sigma_t \cdot \sigma_f \geq \frac{1}{4\pi} - \text{you cannot have both.}$$

	Time res.	Freq. res.
Long window	poor	good
Short window	good	poor
Morlet wavelet	adaptive	adaptive

Morlet wavelet: same *relative* bandwidth at all frequencies - the natural choice for neural data spanning δ to γ .

$$P_{\text{norm}}(f, t) = 10 \log_{10} \left(\frac{P(f, t)}{P_{\text{bl}}(f)} \right)$$

$P_{\text{bl}}(f)$ median power over baseline window

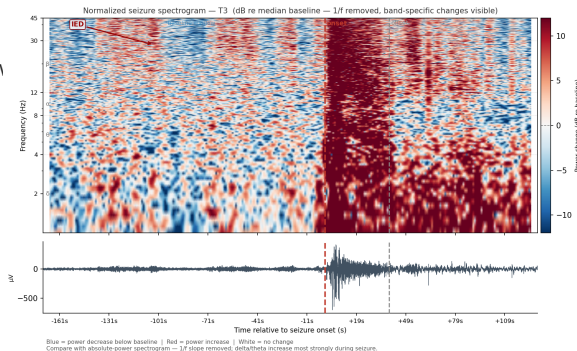
0 dB no change from baseline

+dB power increase

-dB power decrease

Baseline window (BL_START = 220,
BL_END = 360):

- Must be clean (no artifacts, no pre-ictal activity)
- Far enough from seizure onset (>60 s)
- Using the 60 s *immediately* before onset risks capturing pre-ictal changes



dB re median baseline. Channel T3. Same recording as Section 5.

Standard EEG bands

Band	Hz	Association
δ	0.5-4	sleep, post-ictal
θ	4-8	memory, language
α	8-12	idle, eyes closed
β	13-30	cognition, motor
γ	30-45	HFOs, processing

Welch: $n_{\text{perseg}} = 4 \text{ s} \Rightarrow \Delta f = 0.25 \text{ Hz}$

Min resolvable $f = 1/\text{epoch_duration}$

Epochs shorter than n_{perseg} (4s):

delta/theta estimates unreliable.

For the 66 s seizure in aaaaakbz: $\min f \approx 0.015 \text{ Hz}$ - well below delta. Welch at 4 s segments needs $>4 \text{ s}$ of data; delta/theta are reliable here.

Expected ictal changes (cpsz)

Broadband increase - hypersynchronous recruitment.

Band	Ictal vs. pre	Post-ictal
δ	$\uparrow\uparrow$ large	$\uparrow$ persists
θ	$\uparrow\uparrow$ large	$\approx$ baseline
α	$\uparrow$ / suppressed	$\approx$ baseline
β	$\uparrow$	$\approx$ baseline
γ	$\uparrow$ small	slight $\downarrow$

Post-ictal δ slowing = cortical inhibition; confirms seizure just occurred.

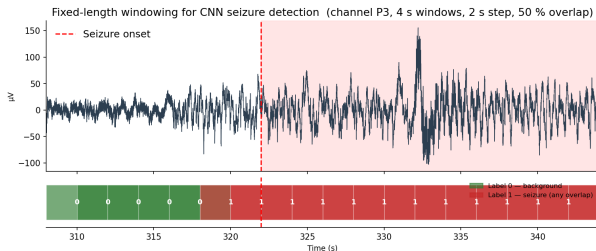
Parameters

Window 4 s ($\text{WIN_S} = 4.0$)
Hop 2 s - 50 % overlap
Label 1 if $\geq 50\%$ overlaps seizure

Class imbalance

Seizures are brief events ($<5\%$ of most recordings).

A model predicting “background” always achieves $>95\%$ accuracy but 0 % sensitivity.



TUSZ aaaaasgd, 4 s windows, 50 % overlap. Red = seizure label.

Label threshold matters: “any overlap = seizure” gives more positives but noisier boundaries. $\geq 50\%$ is a common compromise.

Fixes: Class weights in loss function, Weighted random sampler, Lower decision threshold

The training set above only has fnsz and gnsz. Four more types are available locally: absz, cpsz, tcsz, tnsz - see “TUSZ Patients Used in This Workshop”.

Try it

- Add windows from the 4 new patients to training - does sensitivity on fnsz/gnsz improve or degrade?
- Hold one type out entirely (e.g. train with no tcsz windows) - can the model still detect it? Tests true generalization, not memorized morphology.
- Report **sensitivity per type**, not just one aggregate number - a model that is 95 % sensitive overall could be 0 % sensitive on absz alone (very different morphology - see the lecture's seizure type gallery).

Same lesson as the class imbalance slide: TUSZ pools several very different seizure types under one binary label. A model that looks good on average can be silently failing on the rarest, most clinically distinct type.